

Question 1

```
install.packages("TSA")
library(TSA)
```



Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependencies 'xts', 'TTR', 'quadprog', 'zoo', 'quantmod', 'leaps', 'locfit', 'tseries'

Attaching package: 'TSA'

The following objects are masked from 'package:stats':

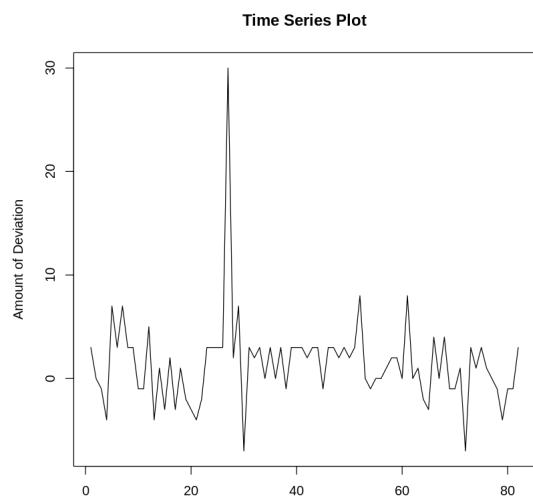
acf, arima

The following object is masked from 'package:utils':

tar

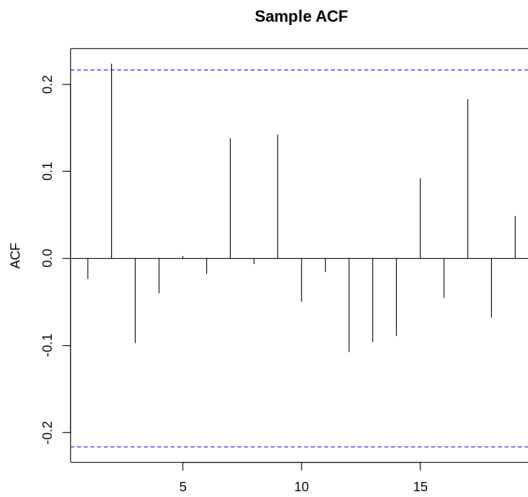
```
data(deere1)
```

```
ts.plot(deere1, main = "Time Series Plot", ylab = "Amount of Deviation")
```



There is a significant spike around 27 time, other than this whole data seems to be within same range with zero at center.

```
acf(deere1, main = "Sample ACF")
```



there is significant autocorrelation at lag 1, meaning not high dependence on previous value, no clear pattern is there in the ACF.

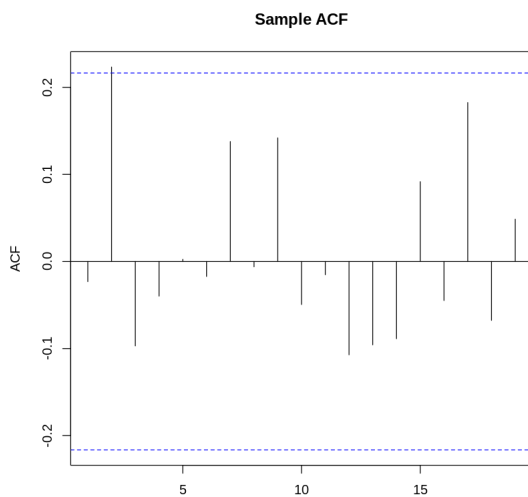
```
which.max(deere1)
which.min(deere1)
```



27

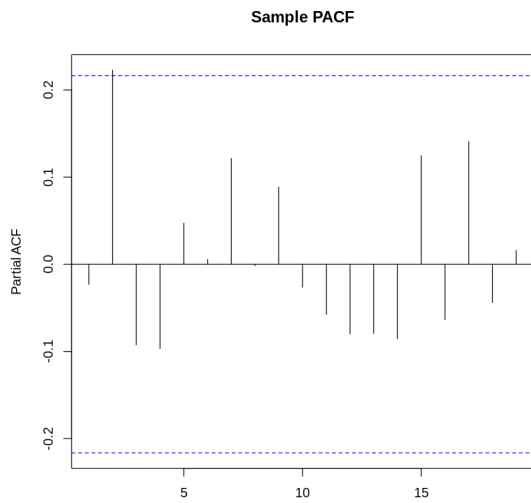
```
#we are replacing the max value which is outlier with the mean of maximum and minimum values in the time series.
mean_max <- mean(deere1[-c(27, 30)])
deere1[27] <- mean_max
```

```
acf(deere1, main = "Sample ACF")
```



now the correlation is more consistent, the lag at 1 remains but for other lags the ACF value decreased, thus meaning outlier is affecting them.

```
pacf(deere1, main = "Sample PACF")
```



```
install.packages("forecast")
```



Installing package into `'/usr/local/lib/R/site-library'`
(as `'lib'` is unspecified)

also installing the dependencies `'fracdiff'`, `'lmtest'`, `'timeDate'`, `'urca'`, `'RcppArmadillo'`

```
library(forecast)
```

```
#I suggest an arima model, as there is no pattern and arima can also capture the dependency we observed even after modifying
```

```
model <- auto.arima(deere1)
```

```
summary(model)
```

```
tsdiag(model)
```



Series: deere1
ARIMA(0,0,0) with non-zero mean

Coefficients:

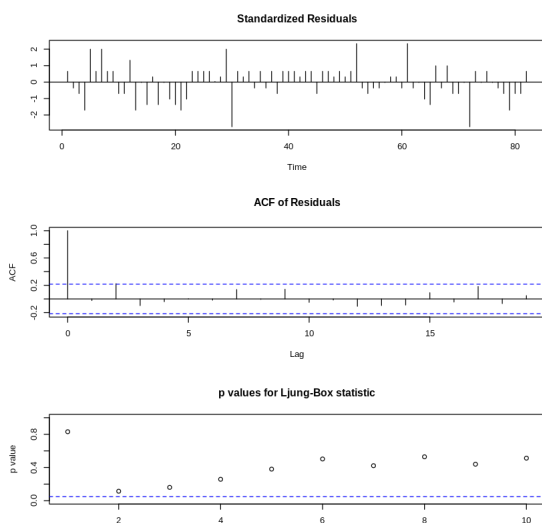
mean
1.0630
s.e. 0.3265

$\sigma^2 = 8.848$: log likelihood = -205.24

AIC=414.48 AICc=414.63 BIC=419.29

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	-3.644673e-13	2.956393	2.33125	-Inf	Inf	0.7563082	-0.02314723



Question 2

```

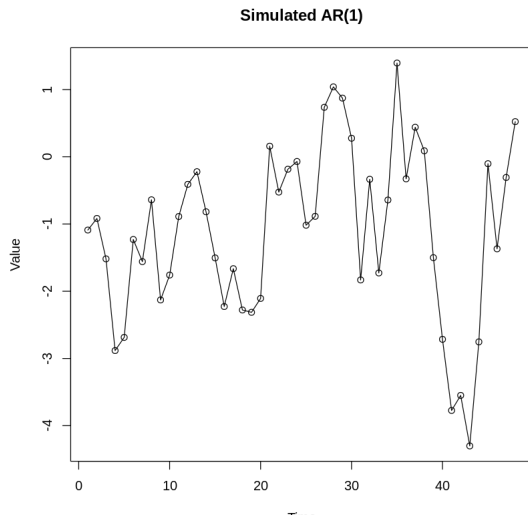
n <- 48
phi <- 0.8
ar1_series <- arima.sim(model = list(ar = phi), n = n)

plot(ar1_series, type = "o", main = "Simulated AR(1)", ylab = "Value", xlab = "Time")

num <- sum(ar1_series[-1] * ar1_series[-n])
den <- sum(ar1_series[-n]^2)
phi_mm <- num / den
phi_mm

```

0.788976593474393



```

phi_cls <- phi_mm
phi_cls #conditional least squares estimate of phi

```

```

mle <- arima(ar1_series, order = c(1, 0, 0), method = "ML")
phi_mle <- mle$coef["ar1"]
phi_mle

```

```

ar1_series_new <- arima.sim(model = list(ar = phi), n = n)

plot(ar1_series_new, type = "o", main = "New Simulated AR(1)", ylab = "Value", xlab = "Time")

num_new <- sum(ar1_series_new[-1] * ar1_series_new[-n])
den_new <- sum(ar1_series_new[-n]^2)
phi_mm_new <- num_new / den_new

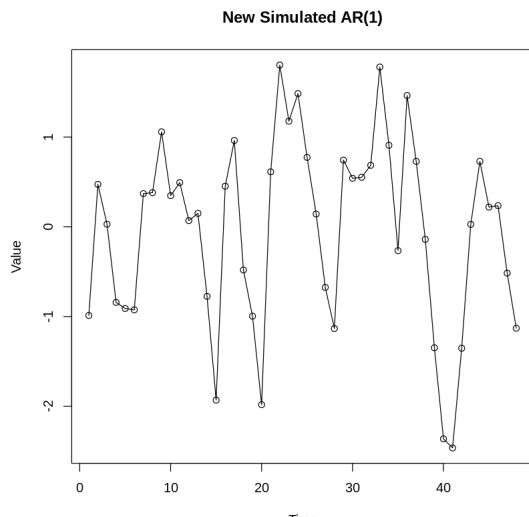
phi_cls_new <- phi_mm_new

mle_model_new <- arima(ar1_series_new, order = c(1, 0, 0), method = "ML")
phi_mle_new <- mle_model_new$coef["ar1"]

phi_mm_new
phi_cls_new
phi_mle_new

```

```
0.540910416482701
0.540910416482701
ar1: 0.539978016841171
```



Question 3

```
library(TSA)
data(robot)

ar1_model <- arima(robot, order = c(1, 0, 0))
summary(ar1_model)

aic_ar1 <- ar1_model$aic
aic_ar1
```

Warning message in trainingaccuracy(object, test, d, D):
"test elements must be within sample"

Call:
arima(x = robot, order = c(1, 0, 0))

Coefficients:

	ar1	intercept
	0.3074	0.0015
s.e.	0.0528	0.0002

sigma^2 estimated as 6.482e-06: log likelihood = 1475.54, aic = -2947.08

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE
Training set	NaN	NaN	NaN	NaN	NaN

```
ima <- arima(robot, order = c(0, 1, 1))
summary(ima)
```

```
aic_ima <- ima$aic
aic_ima
```

Warning message in trainingaccuracy(object, test, d, D):
"test elements must be within sample"

Call:
arima(x = robot, order = c(0, 1, 1))

Coefficients:

	ma1
	-0.8713
s.e.	0.0389

sigma^2 estimated as 6.069e-06: log likelihood = 1480.95, aic = -2959.9

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE
Training set	NaN	NaN	NaN	NaN	NaN

IMA(1,1) model (AIC = -2959.90) performs better than the AR(1) model (AIC = -2947.08). IMA(1,1) model works better because of its differencing and moving average component which were absent in the other model.

Question 4

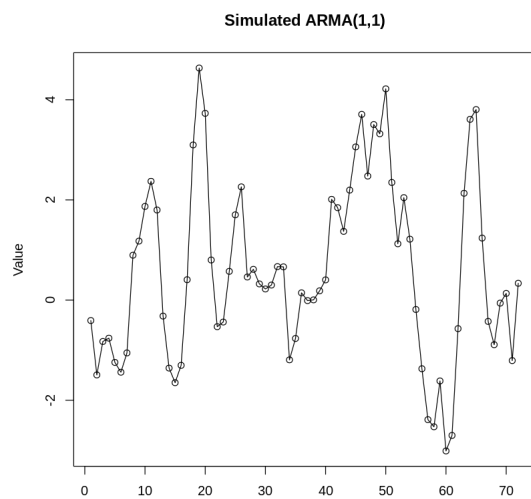
```
n <- 72
phi <- 0.7
theta <- 0.4
arma_series <- arima.sim(model = list(ar = phi, ma = theta), n = n)

plot(arma_series, type = "o", main = "Simulated ARMA(1,1)", ylab = "Value", xlab = "Time")

acf_vals <- acf(arma_series, plot = FALSE)
rho_1 <- acf_vals$acf[2]
rho_2 <- acf_vals$acf[3]

phi_mm <- rho_1 * (1 + theta) / (1 + theta * rho_1)
theta_mm <- (rho_1 - phi_mm) / (1 - phi_mm)
phi_mm
theta_mm
```

```
0.507426042379141
-0.169561738344656
```



```
cls <- arima(arma_series, order = c(1, 0, 1), method = "CSS")
phi_cls <- cls$coef["ar1"]
theta_cls <- cls$coef["ma1"]
phi_cls
theta_cls
```

```
ar1: 0.656665117121453
```

```
mle <- arima(arma_series, order = c(1, 0, 1), method = "ML")
phi_mle <- mle$coef["ar1"]
theta_mle <- mle$coef["ma1"]
phi_mle
theta_mle
```

```
ar1: 0.638302872849366
```

```
arma_series_new <- arima.sim(model = list(ar = phi, ma = theta), n = n)

acf_vals_new <- acf(arma_series_new, plot = FALSE)
rho_1_new <- acf_vals_new$acf[2]
rho_2_new <- acf_vals_new$acf[3]

phi_mm_new <- rho_1_new * (1 + theta) / (1 + theta * rho_1_new)
theta_mm_new <- (rho_1_new - phi_mm_new) / (1 - phi_mm_new)

cls_new <- arima(arma_series_new, order = c(1, 0, 1), method = "CSS")
phi_cls_new <- cls_new$coef["ar1"]
theta_cls_new <- cls_new$coef["ma1"]
```

```
mle_new <- arima(arma_series_new, order = c(1, 0, 1), method = "ML")  
phi_mle_new <- mle_new$coef["ar1"]  
theta_mle_new <- mle_new$coef["ma1"]
```

```
phi_mm_new  
theta_mm_new  
phi_cls_new  
theta_cls_new  
phi_mle_new  
theta_mle_new
```

```
↵ 0.486639757425301  
-0.161494052569534  
ar1: 0.553477682697457  
ma1: 0.399533480893048  
ar1: 0.555607980213967  
ma1: 0.390585955745266
```

The new AR(1) estimates are higher than the old, so new simulations have strong correlation. The MA(1) estimates changed in both sign and magnitude.